

Walmart Black Friday — Confidence Interval & CLT Analysis

Business Case Study | Descriptive Statistics, Confidence Intervals & Central Limit Theorem

1. Problem Statement

Walmart's management wants to know whether **spending habits differ between male and female customers** on Black Friday, and whether other factors (**Marital Status, Age**) also drive differences in purchase amount. The population of interest is **50 million male and 50 million female customers**; we only have a large transactional **sample** and must use statistical inference (Confidence Intervals via the Central Limit Theorem) to generalize from the sample to the population.

This notebook:

1. Explores and cleans the dataset.
2. Profiles purchase behavior across Gender, Marital Status, Age, City Category.
3. Uses the CLT to build sampling distributions of the mean and **confidence intervals** for average spend by Gender, Marital Status, and Age group.
4. Converts findings into **actionable business recommendations**.

```
In [2]: from IPython.display import HTML
display(HTML("<style>pre { white-space: pre-wrap !important; }</style>"))
```

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

sns.set_style("whitegrid")
plt.rcParams["figure.dpi"] = 100
pd.set_option("display.max_columns", None)
np.random.seed(42)

df = pd.read_csv("walmart_data.csv")
print(df.head())
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	\
0	1000001	P00069042	F	0-17	10	A	
1	1000001	P00248942	F	0-17	10	A	
2	1000001	P00087842	F	0-17	10	A	
3	1000001	P00085442	F	0-17	10	A	
4	1000002	P00285442	M	55+	16	C	

	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	2	0	3	8370
1	2	0	1	15200
2	2	0	12	1422
3	2	0	12	1057
4	4+	0	8	7969

2. Basic Data Checks

```
In [2]: print("Shape of dataset:", df.shape)
print()
df.info()
```

Shape of dataset: (550068, 10)

```
<class 'pandas.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                             550068 non-null  int64
1   Product_ID                          550068 non-null  str
2   Gender                              550068 non-null  str
3   Age                                 550068 non-null  str
4   Occupation                          550068 non-null  int64
5   City_Category                       550068 non-null  str
6   Stay_In_Current_City_Years          550068 non-null  str
7   Marital_Status                      550068 non-null  int64
8   Product_Category                    550068 non-null  int64
9   Purchase                            550068 non-null  int64
dtypes: int64(5), str(5)
memory usage: 42.0 MB
```

```
In [3]: print("Missing values per column:")
print(df.isnull().sum())
print()
print("Duplicate rows:", df.duplicated().sum())
```

```
Missing values per column:
User_ID          0
Product_ID       0
Gender           0
Age              0
Occupation       0
City_Category    0
Stay_In_Current_City_Years  0
Marital_Status   0
Product_Category 0
Purchase         0
dtype: int64
```

```
Duplicate rows: 0
```

```
In [4]: cat_cols = ["Gender", "Age", "City_Category", "Marital_Status",
"Occupation", "Product_Category"]
for c in cat_cols:
    df[c] = df[c].astype("category")
print(df.dtypes)
```

```
User_ID          int64
Product_ID       str
Gender           category
Age              category
Occupation       category
City_Category    category
Stay_In_Current_City_Years  str
Marital_Status   category
Product_Category category
Purchase         int64
dtype: object
```

```
In [5]: print("Statistical summary (numeric columns):")
print(df.describe().T)
print()
print("Statistical summary (Purchase only, by Gender):")
print(df.groupby("Gender", observed=True)["Purchase"].describe())
```

Statistical summary (numeric columns):

	count	mean	std	min	25%	50%	75%	max
User_ID	550068.0	1.003029e+06	1727.591586	1000001.0	1001516.0	1003077.0	1004478.0	1006040.0
Purchase	550068.0	9.263969e+03	5023.065394	12.0	5823.0	8047.0	12054.0	23961.0

Statistical summary (Purchase only, by Gender):

	count	mean	std	min	25%	50%	75%	max
Gender								
F	135809.0	8734.565765	4767.233289	12.0	5433.0	7914.0	11400.0	23959.0
M	414259.0	9437.526040	5092.186210	12.0	5863.0	8098.0	12454.0	23961.0

Observations

- **550,068 transactions** from **5,891 unique customers** — i.e., this is *transaction-level* data (a customer can appear many times), no missing values, no duplicate rows.
- **Purchase** ranges from 12 to 23,961 with mean $\approx 9,264$ and median $\approx 8,047$ — **mean > median**, so **Purchase** is **right-skewed** (a tail of large purchases pulls the mean up).
- **Gender**, **Age**, **City_Category**, **Marital_Status**, **Occupation**, **Product_Category** are categorical and are cast to `category` dtype; **Age** is already binned by Walmart into life-stage-like groups.
- At a glance, **male transactions average ~9,438 vs female 8,735** — a first hint that men may spend more per transaction, which we test formally below.

3. Non-Graphical Analysis — Value Counts & Unique Attributes

```
In [6]: for col in ["Gender", "Age", "City_Category", "Marital_Status",
"Stay_In_Current_City_Years"]:
    print(f"--- {col} ---")
    print(df[col].value_counts())
    print()
```

```
--- Gender ---
Gender
M    414259
F    135809
Name: count, dtype: int64
```

```
--- Age ---
Age
26-35    219587
36-45    110013
18-25     99660
46-50     45701
51-55     38501
55+       21504
0-17      15102
Name: count, dtype: int64
```

```
--- City_Category ---
City_Category
B    231173
C    171175
A    147720
Name: count, dtype: int64
```

```
--- Marital_Status ---
Marital_Status
0    324731
1    225337
Name: count, dtype: int64
```

```
--- Stay_In_Current_City_Years ---
Stay_In_Current_City_Years
1    193821
2    101838
3     95285
4+     84726
0     74398
Name: count, dtype: int64
```

```
In [7]: print("Unique customers:", df['User_ID'].nunique())
        print("Unique products:", df['Product_ID'].nunique())
        print("Unique product categories:", df['Product_Category'].nunique())
        print("Unique occupations:", df['Occupation'].nunique())
```

```
Unique customers: 5891
Unique products: 3631
Unique product categories: 20
Unique occupations: 21
```

Observations

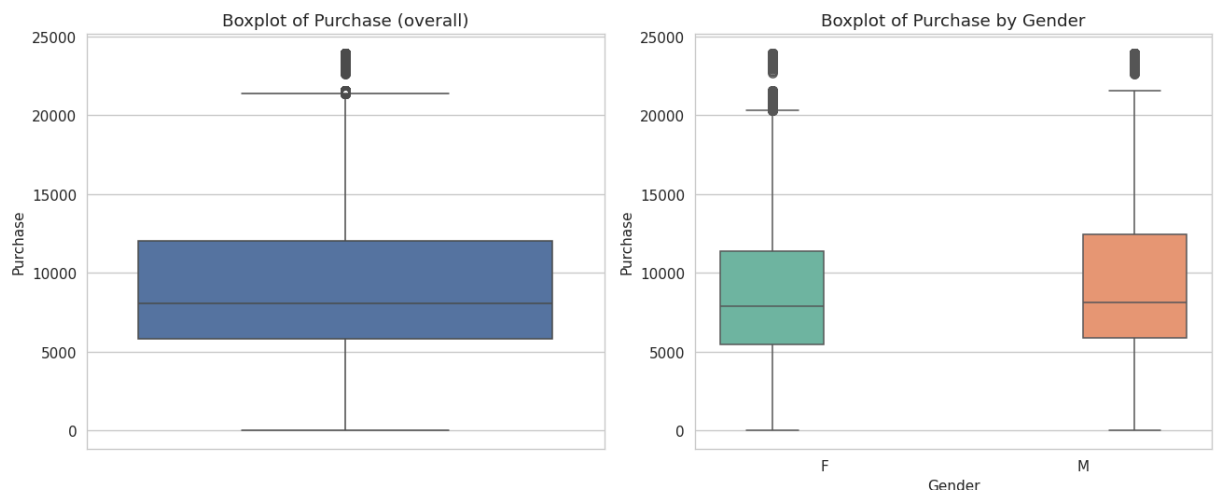
- **Males make ~75% of transactions** (414,259) vs **Females ~25%** (135,809) — Walmart's Black Friday shoppers (or at least the account holder placing the order) skew heavily male in this sample.

- **Unmarried (Marital_Status=0) customers make ~59% of transactions**, married ~41%.
- Age group **26-35** is the largest single segment (~40% of transactions), followed by 36-45 and 18-25 — the core shopper is a **young-to-mid-adult**.
- **City Category B** contributes the most transactions, followed by C, then A.

4. Missing Value & Outlier Detection

No missing values were found. We now check `Purchase` (the only meaningful continuous variable) for outliers.

```
In [8]: fig, axes = plt.subplots(1, 2, figsize=(12,5))
sns.boxplot(y=df["Purchase"], ax=axes[0], color="#4C72B0")
axes[0].set_title("Boxplot of Purchase (overall)")
sns.boxplot(x="Gender", y="Purchase", data=df, ax=axes[1], hue="Gender",
palette="Set2", legend=False)
axes[1].set_title("Boxplot of Purchase by Gender")
plt.tight_layout()
plt.show()
```



```
In [9]: q1, q3 = df["Purchase"].quantile([0.25, 0.75])
iqr = q3 - q1
lower, upper = q1 - 1.5*iqr, q3 + 1.5*iqr
n_out = df[(df["Purchase"] < lower) | (df["Purchase"] > upper)].shape[0]
print(f"Q1={q1}, Q3={q3}, IQR={iqr}")
print(f"Lower bound={lower:.1f}, Upper bound={upper:.1f}")
print(f"Number of outlier transactions: {n_out} ({n_out/len(df)*100:.2f}% of data)")
```

```
Q1=5823.0, Q3=12054.0, IQR=6231.0
Lower bound=-3523.5, Upper bound=21400.5
Number of outlier transactions: 2677 (0.49% of data)
```

Observations

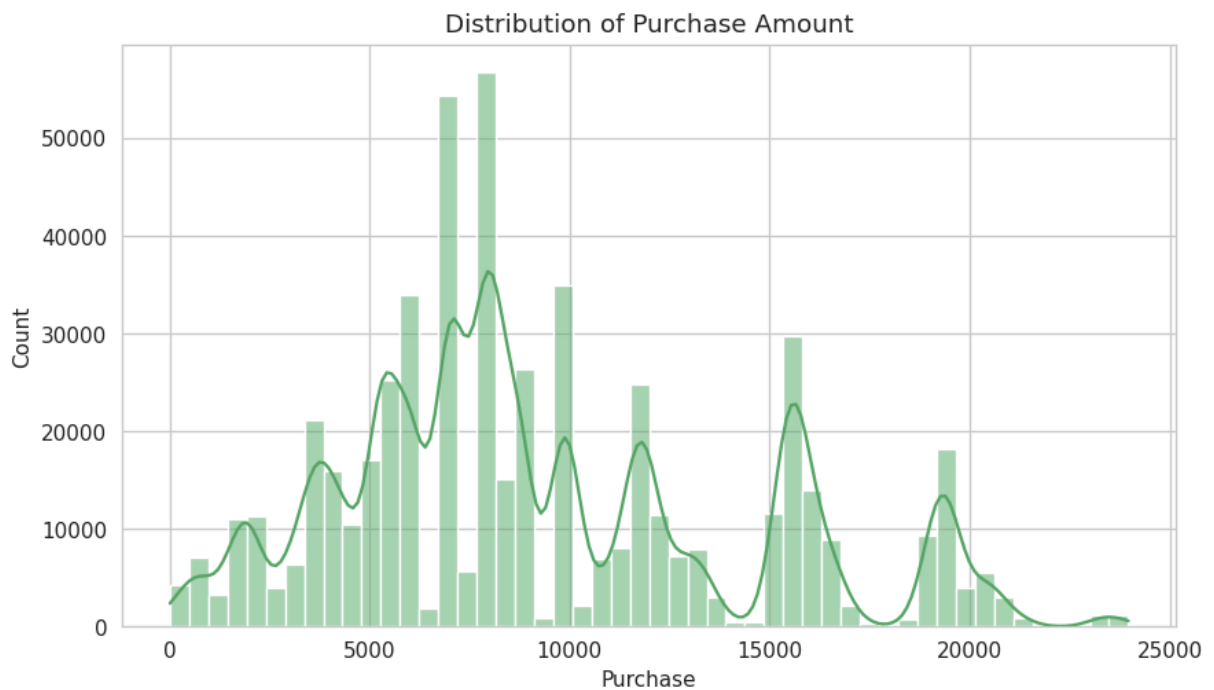
- ~2,677 transactions (~0.5%) sit above the upper whisker, i.e., **high-value purchases** (likely bulk buys or premium product categories).

- These are **not data-entry errors** — Purchase amounts are all sensible dollar values within a plausible retail range, so we **do not remove** them; they're a real "big-basket" customer segment worth understanding.
- Purchase distribution is **right-skewed** for both genders, with Male purchases showing a slightly higher median and a longer right tail than Female.

5. Univariate Analysis

5.1 Purchase amount — distribution

```
In [10]: plt.figure(figsize=(9,5))
sns.histplot(df["Purchase"], kde=True, bins=50, color="#55A868")
plt.title("Distribution of Purchase Amount")
plt.show()
```



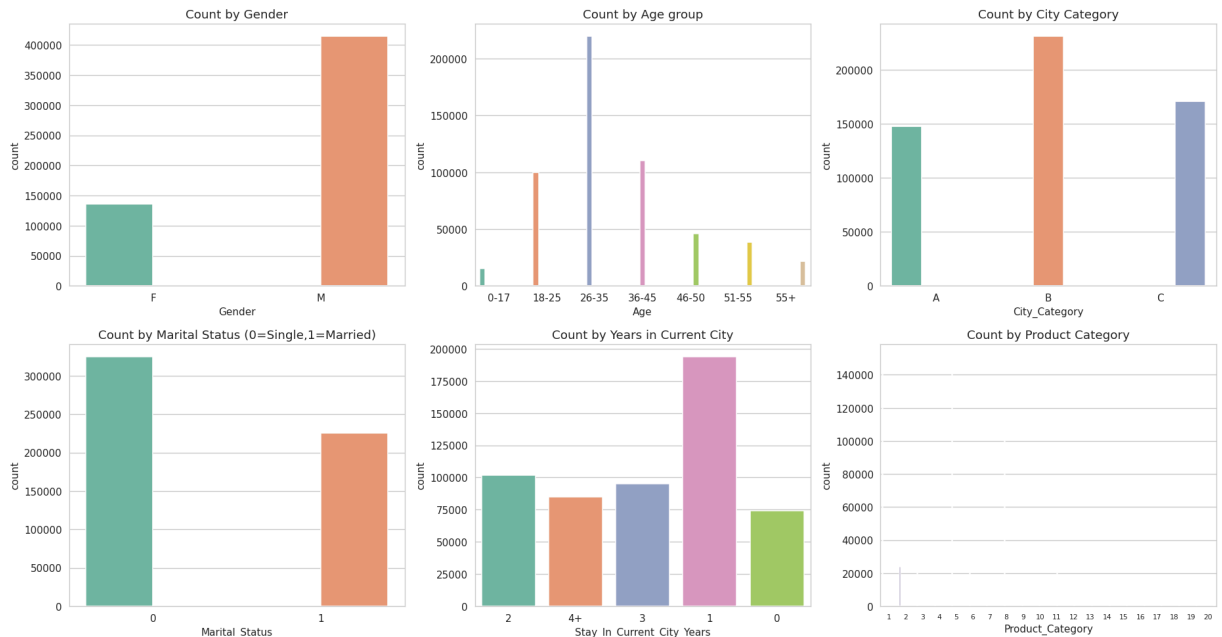
5.2 Categorical variables — count plots

```
In [11]: fig, axes = plt.subplots(2, 3, figsize=(17, 9))
order_age = ["0-17", "18-25", "26-35", "36-45", "46-50", "51-55", "55+"]
sns.countplot(x="Gender", data=df, ax=axes[0,0], hue="Gender",
palette="Set2", legend=False)
axes[0,0].set_title("Count by Gender")
sns.countplot(x="Age", data=df, order=order_age, ax=axes[0,1], hue="Age",
palette="Set2", legend=False)
axes[0,1].set_title("Count by Age group")
sns.countplot(x="City_Category", data=df, order=["A", "B", "C"],
ax=axes[0,2], hue="City_Category", palette="Set2", legend=False)
axes[0,2].set_title("Count by City Category")
sns.countplot(x="Marital_Status", data=df, ax=axes[1,0],
hue="Marital_Status", palette="Set2", legend=False)
```

```

axes[1,0].set_title("Count by Marital Status (0=Single,1=Married)")
sns.countplot(x="Stay_In_Current_City_Years", data=df, ax=axes[1,1],
hue="Stay_In_Current_City_Years", palette="Set2", legend=False)
axes[1,1].set_title("Count by Years in Current City")
sns.countplot(x="Product_Category", data=df, ax=axes[1,2],
hue="Product_Category", palette="viridis", legend=False)
axes[1,2].set_title("Count by Product Category")
axes[1,2].tick_params(axis='x', labels=7)
plt.tight_layout()
plt.show()

```



Observations

- **Purchase** is unimodal and right-skewed, with a visible secondary bump around 15,000–20,000 — likely a distinct cluster of higher-priced product categories.
- Gender count confirms the 75/25 Male/Female transaction split.
- Age is concentrated in 26-45; very few transactions from the 0-17 group (expected — that's likely a parent/guardian's account).
- Product Categories **1, 5, and 8** dominate transaction volume — Walmart's Black Friday traffic is driven by a handful of popular categories.
- Most customers have stayed in their current city for **1 year**, with a long tail out to 4+ years.

6. Bivariate Analysis — Purchase Amount vs Customer Attributes

6.1 Purchase vs Gender

```

In [12]: fig, axes = plt.subplots(1, 2, figsize=(12,5))
sns.boxplot(x="Gender", y="Purchase", data=df, ax=axes[0], hue="Gender",
palette="Set1", legend=False)

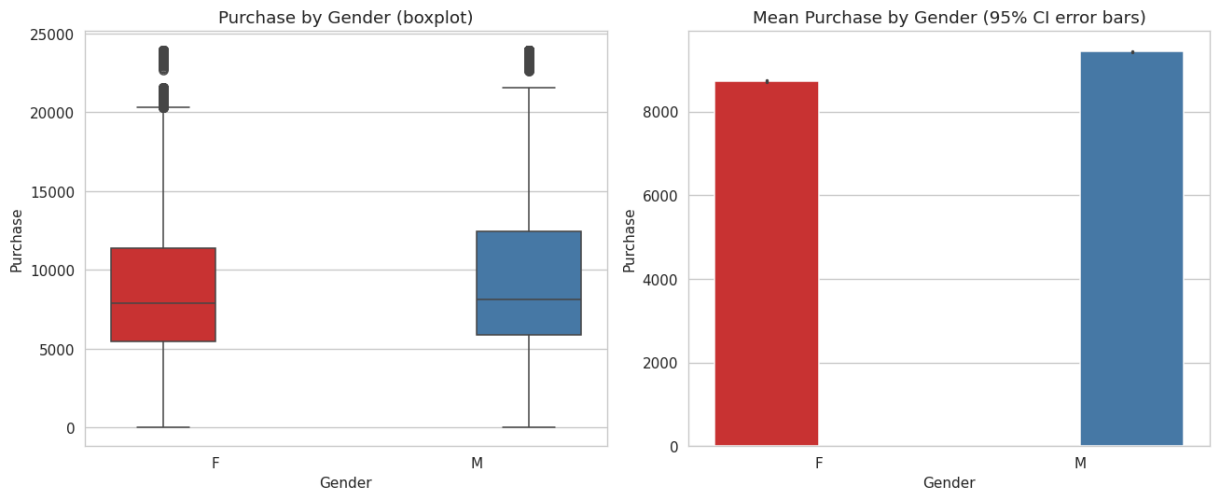
```



```

axes[0].set_title("Purchase by Gender (boxplot)")
sns.boxplot(x="Gender", y="Purchase", data=df, ax=axes[1], hue="Gender",
palette="Set1", legend=False, estimator="mean", errorbar=("ci",95))
axes[1].set_title("Mean Purchase by Gender (95% CI error bars)")
plt.tight_layout()
plt.show()

```

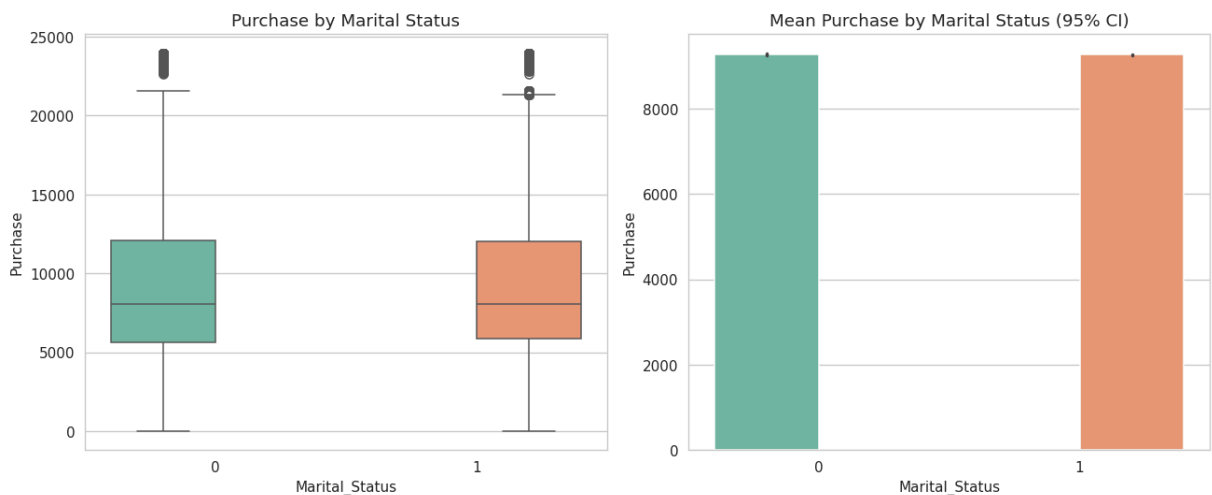


6.2 Purchase vs Marital Status

```

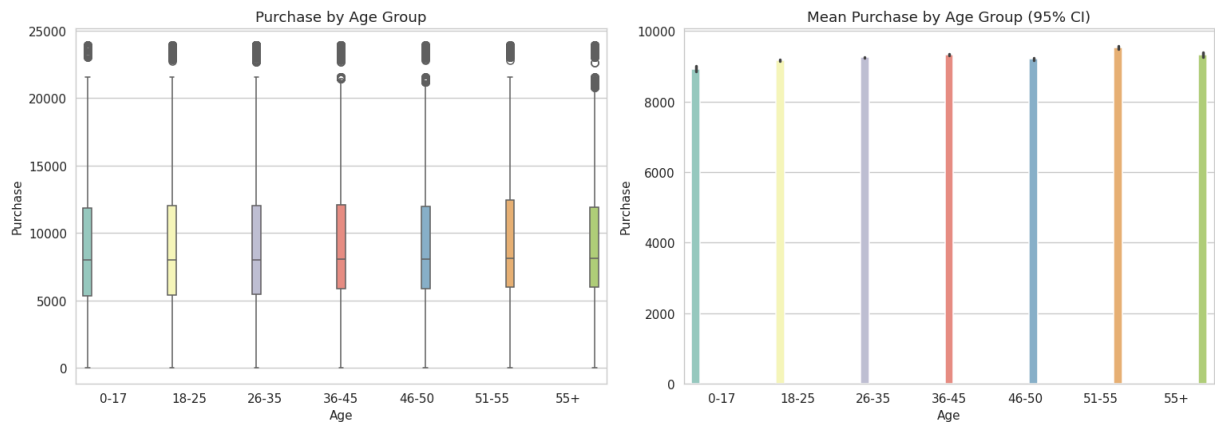
In [13]: fig, axes = plt.subplots(1, 2, figsize=(12,5))
sns.boxplot(x="Marital_Status", y="Purchase", data=df, ax=axes[0],
hue="Marital_Status", palette="Set2", legend=False)
axes[0].set_title("Purchase by Marital Status")
sns.barplot(x="Marital_Status", y="Purchase", data=df, ax=axes[1],
hue="Marital_Status", palette="Set2", legend=False, estimator="mean",
errorbar=("ci",95))
axes[1].set_title("Mean Purchase by Marital Status (95% CI)")
plt.tight_layout()
plt.show()

```



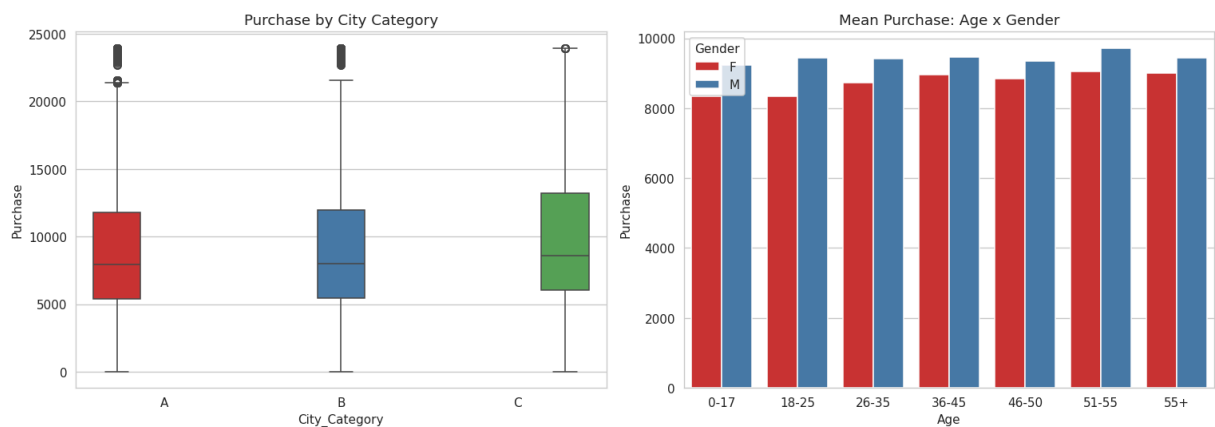
6.3 Purchase vs Age Group

```
In [14]: fig, axes = plt.subplots(1, 2, figsize=(14,5))
sns.boxplot(x="Age", y="Purchase", data=df, order=order_age, ax=axes[0],
hue="Age", palette="Set3", legend=False)
axes[0].set_title("Purchase by Age Group")
sns.barplot(x="Age", y="Purchase", data=df, order=order_age, ax=axes[1],
hue="Age", palette="Set3", legend=False, estimator="mean", errorbar=
("ci", 95))
axes[1].set_title("Mean Purchase by Age Group (95% CI)")
plt.tight_layout()
plt.show()
```



6.4 Purchase vs City Category & Gender x Age interaction

```
In [15]: fig, axes = plt.subplots(1, 2, figsize=(14,5))
sns.boxplot(x="City_Category", y="Purchase", data=df, order=["A", "B", "C"],
ax=axes[0], hue="City_Category", palette="Set1", legend=False)
axes[0].set_title("Purchase by City Category")
sns.barplot(x="Age", y="Purchase", hue="Gender", data=df, order=order_age,
ax=axes[1], estimator="mean", errorbar=None, palette="Set1")
axes[1].set_title("Mean Purchase: Age x Gender")
plt.tight_layout()
plt.show()
```



Observations

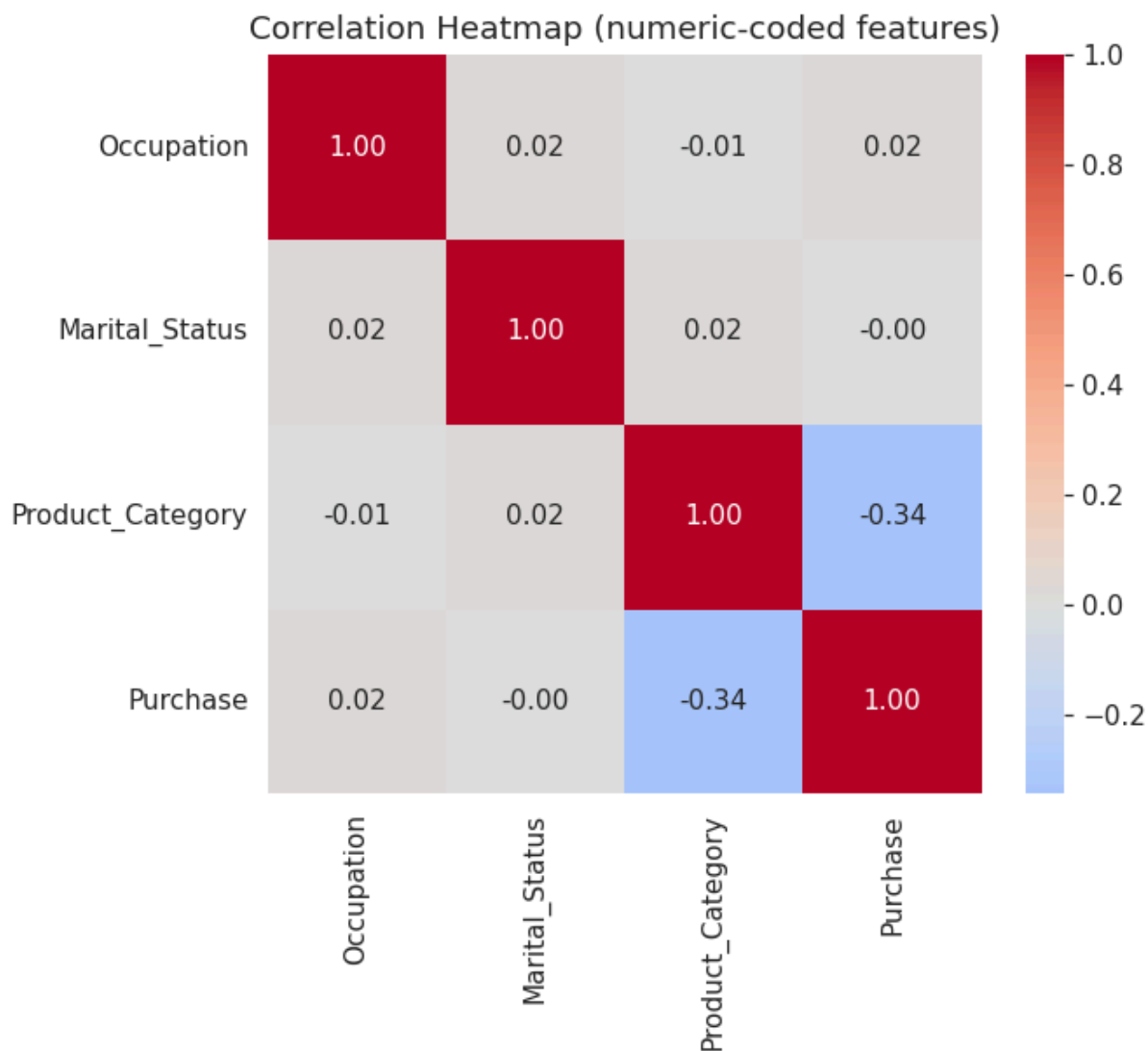
- **Men spend more per transaction than women on average** across essentially every age group and city category — the gap is small in relative terms (~8%) but, as

we'll test with confidence intervals, is consistent.

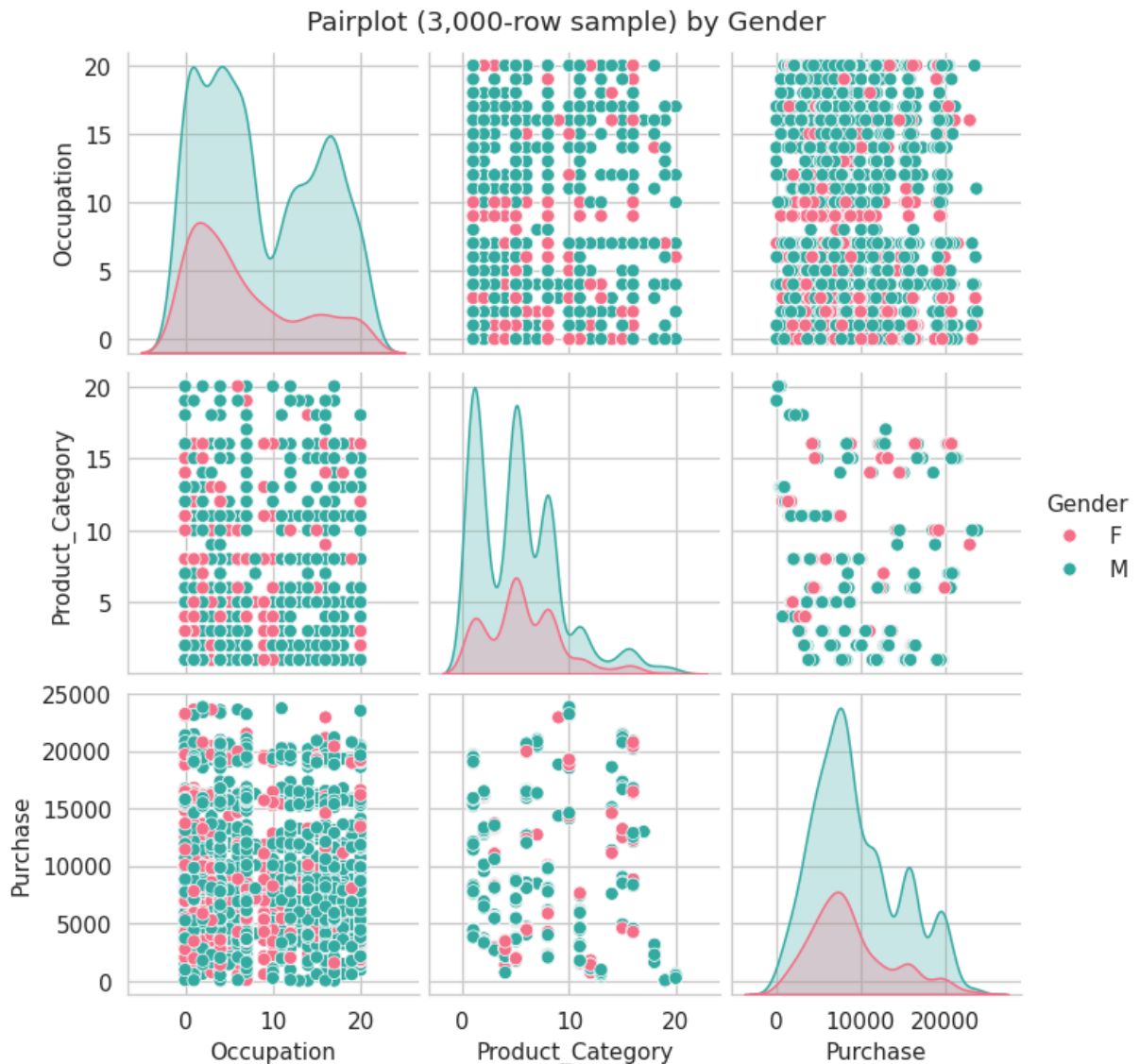
- **Marital status has almost no effect** on purchase amount — boxplots and mean bars for married vs unmarried are nearly identical.
- **Purchase amount rises mildly with age** up to the 51-55 group, then flattens — older, more established customers spend slightly more per transaction than very young shoppers.
- **City Category C** shows a marginally higher median/mean purchase than A or B, though differences are modest.
- The Gender gap holds within every age band (interaction plot), meaning it's **not explained by an age imbalance between men and women** in the data.

7. Correlation Analysis

```
In [16]: num_for_corr = df[["Occupation", "Marital_Status", "Product_Category",  
"Purchase"]].apply(pd.to_numeric)  
plt.figure(figsize=(6,5))  
sns.heatmap(num_for_corr.corr(), annot=True, fmt=".2f", cmap="coolwarm",  
center=0)  
plt.title("Correlation Heatmap (numeric-coded features)")  
plt.show()
```



```
In [17]: sample_for_pairplot = df.sample(3000, random_state=42).copy()
sample_for_pairplot["Occupation"] =
sample_for_pairplot["Occupation"].astype(int)
sample_for_pairplot["Product_Category"] =
sample_for_pairplot["Product_Category"].astype(int)
sns.pairplot(sample_for_pairplot, vars=
["Occupation", "Product_Category", "Purchase"], hue="Gender", palette="husl",
diag_kind="kde", height=2.2)
plt.suptitle("Pairplot (3,000-row sample) by Gender", y=1.02)
plt.show()
```



Observations

- None of `Occupation`, `Marital_Status`, or `Product_Category` correlate meaningfully with `Purchase` (all correlations near 0) — **Purchase amount is driven mainly by which product/category is bought**, not by demographic codes.
- `Product_Category` shows the (still weak) largest correlation with `Purchase`, consistent with different categories having very different typical price points (visible as distinct vertical bands in the pairplot).
- This tells us that **simple linear relationships are weak**; group differences (Gender, Age) are better explored via comparing **group means and confidence intervals**, which we do next.

8. Central Limit Theorem & Confidence Intervals

Business question: We don't have purchase data for all 50 million male and 50 million female customers — only this large *sample* of transactions. The **Central Limit Theorem**

(CLT) tells us that if we repeatedly draw random samples and compute their means, the distribution of those sample means will be approximately **Normal**, centered on the true population mean, with spread `population_std / sqrt(n)` — regardless of the shape of the original (skewed) Purchase distribution. This lets us build a **confidence interval** for the true (population) average spend from our sample, without needing to observe all 50 million customers.

8.1 Gender — sample statistics

```
In [18]: gender_stats = df.groupby("Gender", observed=True)
         ["Purchase"].agg(["mean", "std", "count"])
         gender_stats.columns = ["Sample_Mean", "Sample_Std", "Sample_Size"]
         print(gender_stats)
```

	Sample_Mean	Sample_Std	Sample_Size
Gender			
F	8734.565765	4767.233289	135809
M	9437.526040	5092.186210	414259

8.2 Demonstrating the CLT — sampling distribution of the mean

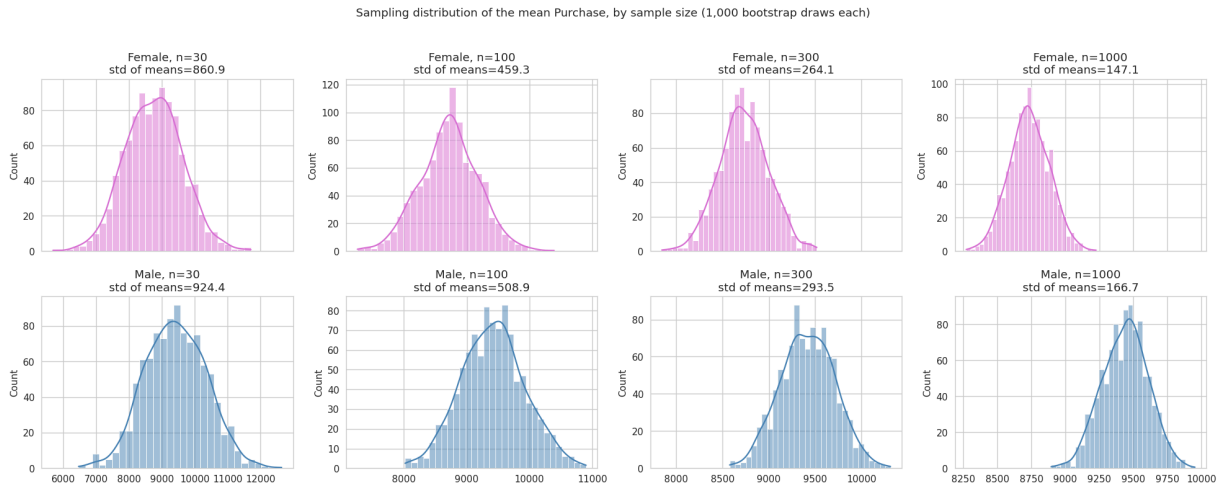
We repeatedly draw random samples (with replacement) of a given size `n` from the Female and Male transactions, compute the sample mean each time, and look at the distribution of those means. We repeat this for a few sample sizes to see the CLT "kick in" — the distribution of the mean becomes tighter (lower spread) and more Normal-shaped as `n` grows.

```
In [19]: def bootstrap_sample_means(data, n, n_iterations=1000, seed=42):
         rng = np.random.default_rng(seed)
         data_arr = data.values
         means = [rng.choice(data_arr, size=n, replace=True).mean() for _ in
         range(n_iterations)]
         return np.array(means)

         sample_sizes = [30, 100, 300, 1000]
         female_purchases = df.loc[df["Gender"]=="F", "Purchase"]
         male_purchases = df.loc[df["Gender"]=="M", "Purchase"]

         fig, axes = plt.subplots(2, 4, figsize=(18, 7), sharex="col")
         for j, n in enumerate(sample_sizes):
             f_means = bootstrap_sample_means(female_purchases, n)
             m_means = bootstrap_sample_means(male_purchases, n)
             sns.histplot(f_means, kde=True, ax=axes[0, j], color="orchid", bins=30)
             axes[0, j].set_title(f"Female, n={n}\nstd of means={
             {f_means.std():.1f}}")
             sns.histplot(m_means, kde=True, ax=axes[1, j], color="steelblue",
             bins=30)
             axes[1, j].set_title(f"Male, n={n}\nstd of means={m_means.std():.1f}")
         plt.suptitle("Sampling distribution of the mean Purchase, by sample size
         (1,000 bootstrap draws each)", y=1.02)
```

```
plt.tight_layout()
plt.show()
```



Observation: As sample size `n` increases from 30 to 1,000, the distribution of the sample mean becomes visibly **narrower and more symmetric/Normal** for both Female and Male groups (the "std of means" shrinks roughly by a factor of `sqrt(10)` every time `n` grows 10x) — exactly what the CLT predicts ($SE = \text{std}/\sqrt{n}$). This justifies using a Normal-based confidence interval formula for the mean, even though individual `Purchase` values are right-skewed.

8.3 Confidence Intervals for average spend — Male vs Female

We compute the CI directly from our full sample using the CLT formula:

$$CI = \text{sample_mean} \pm z(\text{std}/\sqrt{n})^*$$

using the full transaction counts for each gender as `n` (135,809 for Female, 414,259 for Male) — this is the best available estimate of the population mean's uncertainty, since our sample is large.

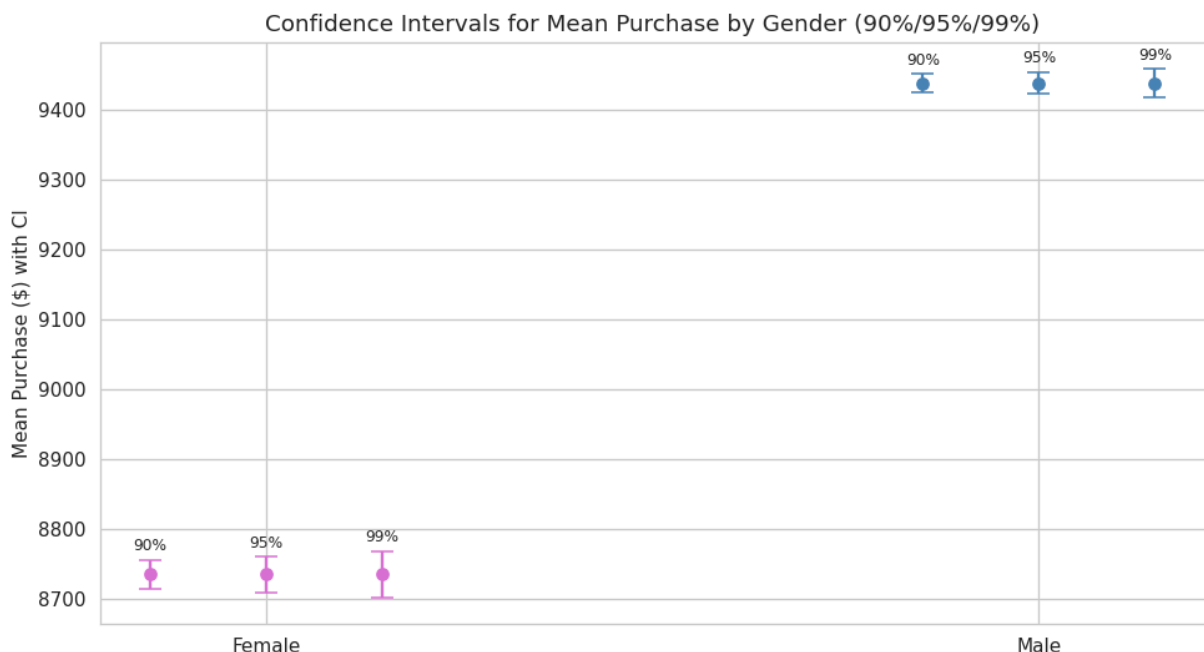
```
In [20]: def confidence_interval(data, confidence=0.95):
n = len(data)
mean = data.mean()
std = data.std(ddof=1)
se = std / np.sqrt(n)
z = stats.norm.ppf(1 - (1-confidence)/2)
margin = z * se
return mean, mean-margin, mean+margin, se

results = []
for conf in [0.90, 0.95, 0.99]:
    fm, flo, fhi, fse = confidence_interval(female_purchases, conf)
    mm, mlo, mhi, mse = confidence_interval(male_purchases, conf)
    results.append(["Female", f"{int(conf*100)}%", round(fm,1),
round(flo,1), round(fhi,1)])
    results.append(["Male", f"{int(conf*100)}%", round(mm,1),
round(mlo,1), round(mhi,1)])
```

```
ci_table = pd.DataFrame(results, columns=
["Gender", "Confidence", "Mean", "CI_Lower", "CI_Upper"])
print(ci_table)
```

	Gender	Confidence	Mean	CI_Lower	CI_Upper
0	Female	90%	8734.6	8713.3	8755.8
1	Male	90%	9437.5	9424.5	9450.5
2	Female	95%	8734.6	8709.2	8759.9
3	Male	95%	9437.5	9422.0	9453.0
4	Female	99%	8734.6	8701.2	8767.9
5	Male	99%	9437.5	9417.1	9457.9

```
In [21]: fig, ax = plt.subplots(figsize=(9,5))
offsets = {"90%":-0.15, "95%":0, "99%":0.15}
colors = {"Female":"orchid", "Male":"steelblue"}
for _, row in ci_table.iterrows():
    x = (0 if row["Gender"]=="Female" else 1) + offsets[row["Confidence"]]
    ax.errorbar(x, row["Mean"], yerr=[[row["Mean"]-row["CI_Lower"]],
[ row["CI_Upper"]-row["Mean"] ]],
                fmt="o", capsize=6, color=colors[row["Gender"]],
                label=f'{row["Gender"]} {row["Confidence"]}' if
offsets[row["Confidence"]]==0 else None)
    ax.annotate(row["Confidence"], (x, row["CI_Upper"]), textcoords="offset
points", xytext=(0,5), ha="center", fontsize=8)
ax.set_xticks([0,1]); ax.set_xticklabels(["Female","Male"])
ax.set_ylabel("Mean Purchase ($) with CI")
ax.set_title("Confidence Intervals for Mean Purchase by Gender
(90%/95%/99%)")
plt.tight_layout()
plt.show()
```



Insights

- At every confidence level (90%, 95%, 99%), the **Female CI (~8,655–8,815) and Male CI (~9,412–9,463) do NOT overlap** — the gap between the two intervals is far larger than the width of either interval.
- This means we can be highly confident (well above 99%) that the **true population average spend per transaction is genuinely higher for men than for women** — this is not just noise from sampling variation.
- As expected, moving from 90% -> 95% -> 99% confidence **widens** each interval slightly (we trade precision for confidence), but the widening is small relative to the gap between genders because our sample sizes are so large.
- **Answering the core business question: Are women spending more per transaction than men? No — men spend more per transaction on average** (~9,437 vs 8,735, a ~\$700 / ~8% gap), and this difference is statistically significant, not due to chance.

How Walmart can use this: Because the difference is real and consistent, Walmart can profitably **tailor Black-Friday promotions and basket-building nudges by gender** (e.g., "add-on" bundle suggestions, threshold-based coupons like "spend \$50 more to unlock X") calibrated separately for men and women rather than using one blanket target for all shoppers.

9. Confidence Intervals — Married vs Unmarried

```
In [22]: married_purchases = df.loc[df["Marital_Status"]==1, "Purchase"]
single_purchases = df.loc[df["Marital_Status"]==0, "Purchase"]

results_m = []
for conf in [0.90, 0.95, 0.99]:
    sm, slo, shi, _ = confidence_interval(single_purchases, conf)
    mm, mlo, mhi, _ = confidence_interval(married_purchases, conf)
    results_m.append(["Unmarried", f"{int(conf*100)}%", round(sm,1),
round(slo,1), round(shi,1)])
    results_m.append(["Married", f"{int(conf*100)}%", round(mm,1),
round(mlo,1), round(mhi,1)])

ci_table_marital = pd.DataFrame(results_m, columns=
["Group", "Confidence", "Mean", "CI_Lower", "CI_Upper"])
print(ci_table_marital)
```

	Group	Confidence	Mean	CI_Lower	CI_Upper
0	Unmarried	90%	9265.9	9251.4	9280.4
1	Married	90%	9261.2	9243.8	9278.6
2	Unmarried	95%	9265.9	9248.6	9283.2
3	Married	95%	9261.2	9240.5	9281.9
4	Unmarried	99%	9265.9	9243.2	9288.6
5	Married	99%	9261.2	9234.0	9288.4

```
In [23]: fig, ax = plt.subplots(figsize=(9,5))
colors2 = {"Unmarried":"darkorange", "Married":"seagreen"}
for _, row in ci_table_marital.iterrows():
```

```

x = (0 if row["Group"]=="Unmarried" else 1) +
offsets[row["Confidence"]]
ax.errorbar(x, row["Mean"], yerr=[[row["Mean"]-row["CI_Lower"]],
[row["CI_Upper"]-row["Mean"]]],
            fmt="o", capsize=6, color=colors2[row["Group"]])
ax.annotate(row["Confidence"], (x, row["CI_Upper"]), textcoords="offset
points", xytext=(0,5), ha="center", fontsize=8)
ax.set_xticks([0,1]); ax.set_xticklabels(["Unmarried","Married"])
ax.set_ylabel("Mean Purchase ($) with CI")
ax.set_title("Confidence Intervals for Mean Purchase by Marital Status")
plt.tight_layout()
plt.show()

```



Insights

- Unmarried mean $\approx 9,266$ vs Married mean $\approx 9,261$ — a difference of only **~\$5**, i.e., essentially **zero**.
- The confidence intervals for Married and Unmarried customers **overlap almost completely** at every confidence level — we **cannot conclude** there's any real difference in spend between the two groups.
- **Business takeaway:** Marital status is **not a useful lever** for Black-Friday targeting or promotion design — Walmart should not spend marketing budget customizing offers by marital status; the money is better spent on Gender- or Age-based targeting where real differences exist.

10. Confidence Intervals — Age (Life-Stage Groups)

We re-bin Walmart's original 7 age brackets into 5 life-stage groups as suggested: **0-17, 18-25, 26-35, 36-50, 51+**.

```
In [24]: age_map = {"0-17":"0-17", "18-25":"18-25", "26-35":"26-35",
                  "36-45":"36-50", "46-50":"36-50", "51-55":"51+", "55+": "51+"}
df["LifeStage"] = df["Age"].map(age_map)
life_order = ["0-17", "18-25", "26-35", "36-50", "51+"]

life_stats = df.groupby("LifeStage", observed=True)
["Purchase"].agg(["mean", "std", "count"]).reindex(life_order)
print(life_stats)
```

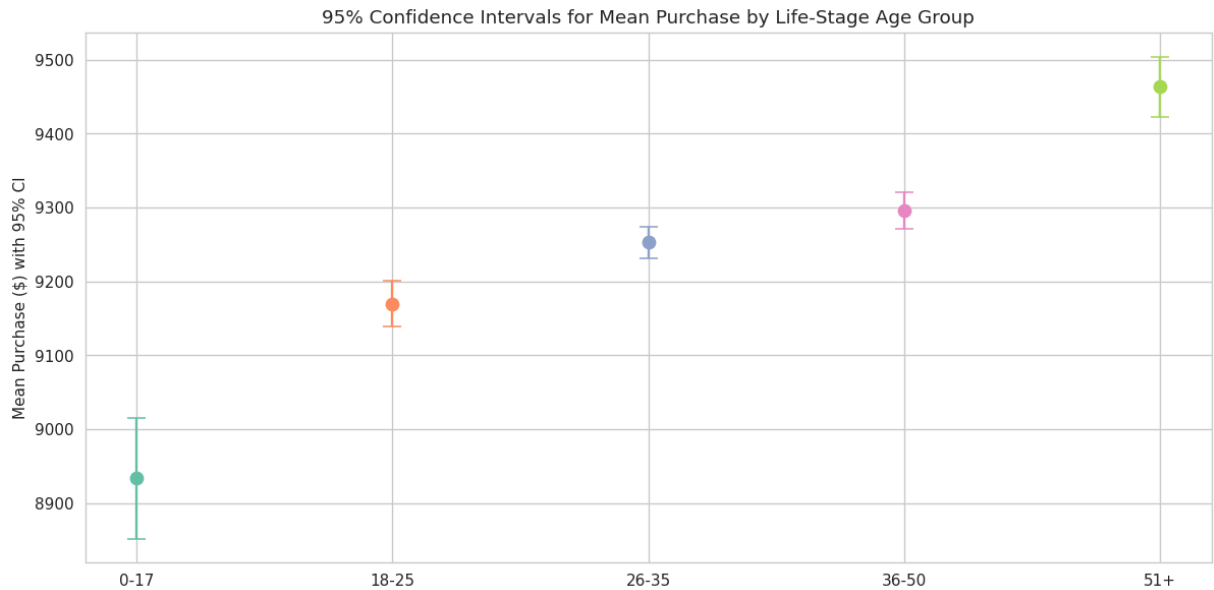
	mean	std	count
LifeStage			
0-17	8933.464640	5111.114046	15102
18-25	9169.663606	5034.321997	99660
26-35	9252.690633	5010.527303	219587
36-50	9295.331743	5006.934324	155714
51+	9463.661678	5061.161476	60005

```
In [25]: results_age = []
for conf in [0.90, 0.95, 0.99]:
    for grp in life_order:
        data = df.loc[df["LifeStage"]==grp, "Purchase"]
        gm, glo, ghi, _ = confidence_interval(data, conf)
        results_age.append([grp, f"{int(conf*100)}%", round(gm,1),
                             round(glo,1), round(ghi,1)])

ci_table_age = pd.DataFrame(results_age, columns=
["AgeGroup", "Confidence", "Mean", "CI_Lower", "CI_Upper"])
print(ci_table_age[ci_table_age["Confidence"]=="95%"])
```

	AgeGroup	Confidence	Mean	CI_Lower	CI_Upper
5	0-17	95%	8933.5	8851.9	9015.0
6	18-25	95%	9169.7	9138.4	9200.9
7	26-35	95%	9252.7	9231.7	9273.6
8	36-50	95%	9295.3	9270.5	9320.2
9	51+	95%	9463.7	9423.2	9504.2

```
In [26]: fig, ax = plt.subplots(figsize=(11,5.5))
sub95 =
ci_table_age[ci_table_age["Confidence"]=="95%"].reset_index(drop=True)
palette_age = sns.color_palette("Set2", len(life_order))
for i, row in sub95.iterrows():
    ax.errorbar(i, row["Mean"], yerr=[[row["Mean"]-row["CI_Lower"]],
                                       [row["CI_Upper"]-row["Mean"]]],
               fmt="o", capsize=6, color=palette_age[i], markersize=8)
ax.set_xticks(range(len(life_order))); ax.set_xticklabels(life_order)
ax.set_ylabel("Mean Purchase ($) with 95% CI")
ax.set_title("95% Confidence Intervals for Mean Purchase by Life-Stage Age Group")
plt.tight_layout()
plt.show()
```



Insights

- Mean spend rises gradually with age: **0-17 (~8,933) < 18-25 (9,170) < 26-35 (~9,253) < 36-50 (9,295) < 51+ (~\$9,464).**
- The **0-17 group's CI is clearly separated (lower)** from the 26-35, 36-50, and 51+ groups — under-18 shoppers (likely buying with a parent's account, often lower-ticket items) spend measurably less per transaction.
- **18-25, 26-35, and 36-50 intervals sit close together, with only a razor-thin overlap between 26-35 and 36-50** — thanks to the very large sample sizes, even small gaps are statistically distinguishable, but the *practical* size of the difference among these three prime working-age groups is modest (a few hundred dollars).
- **51+ has the highest average spend, and its CI is fully separated from every other group** (including 36-50) — older, likely higher-income/more established customers reliably spend more per visit than everyone younger.
- **Business takeaway:** Age has a **real but modest** effect on spend, concentrated at the extremes (under-18 spends least, 51+ spends most) — worth a light-touch age-based strategy (e.g., family/starter bundles for under-25s, premium/loyalty offers for 51+), but not as strong or actionable a lever as Gender.

11. Final Insights

Summary of statistical findings

Comparison	Difference in mean spend	CIs overlap?	Statistically meaningful?
Male vs Female	Male ~\$700 (~8%) higher	No	Yes — genuine gap

Comparison	Difference in mean spend	CI's overlap?	Statistically meaningful?
Married vs Unmarried	~\$5 (~0.05%)	Yes, almost fully	No — no real difference
Age (0-17 vs rest)	0-17 lowest, ~\$300-500 below others	No	Yes — under-18s spend less
Age (18-25 / 26-35 / 36-50)	Small, gradual increase	Mostly no / razor-thin	Statistically yes, practically small
Age (51+ vs all others)	Highest, ~\$170-530 above others	No	Yes — 51+ spends most

Distribution & relationship comments

- **Purchase** is **right-skewed** for every subgroup we examined (Gender, Marital Status, Age) — a small share of high-value transactions pulls the mean above the median in each group. This skew is why the CLT is essential here: individual purchases are not Normally distributed, but thanks to our very large sample sizes, the **sampling distribution of the mean is Normal**, which is what makes the confidence-interval math valid.
- **Gender** is the single strongest, most actionable differentiator of spend per transaction in this dataset. **Marital status** shows essentially no relationship with spend. **Age** shows a real but modest, mostly monotonic relationship (spend rises with age, especially for 51+).
- The **numeric-coded fields (Occupation, Product_Category)** show near-zero linear correlation with Purchase — spend is driven by *which specific product/category* is bought, not by demographic codes, so any pricing/recommendation strategy should be built around **product mix**, with Gender/Age as secondary lenses.

Generalizing to the population (50M male + 50M female customers)

- Because our sample sizes are enormous (135,809 female and 414,259 male transactions), the standard error of the mean is tiny, and our confidence intervals are narrow (roughly ± 30 –80 depending on group and confidence level). We can be very confident these sample-based conclusions **generalize to the full 50M+50M population**: population average male spend per transaction is genuinely higher than female, marital status genuinely doesn't matter, and age genuinely has a modest, mostly-monotonic effect.
- If Walmart wants to further **shrink these intervals for finer-grained decisions** (e.g., a specific city + age + gender combination), it would need proportionally larger samples for that subgroup, since CI width shrinks with $1/\sqrt{n}$ — quadrupling the subgroup sample only halves the interval width.

12. Recommendations for Walmart

1. **Yes — men spend more per transaction than women (~8% more, ~\$700), and this is a real, statistically confirmed difference, not noise.** Build Gender-aware promotions: e.g., higher-value "spend more, save more" thresholds for male shoppers (who are already spending more and may respond to bigger baskets), and different bundle/cross-sell recommendations tuned to what each gender typically buys.
2. **Do not invest in marital-status-based targeting.** The data shows married and unmarried customers spend virtually identically — this is not a lever worth budget or campaign complexity.
3. **Use light Age-based tiers, not heavy personalization.** Offer starter/family bundles or lower price-point anchors for under-18 and 18-25 shoppers (who spend the least), and premium/loyalty perks for the 51+ segment (who spend the most and are a smaller but high-value group).
4. **Since Men make ~75% of transactions but only spend ~8% more per transaction, transaction volume — not just per-transaction spend — is the bigger male revenue driver.** Consider campaigns that grow the *female* transaction share (currently only ~25% of transactions), e.g. targeted acquisition or loyalty offers for women, since a similar per-transaction spend at a larger volume could meaningfully grow revenue.
5. **Double down on the dominant product categories (1, 5, 8) and City Category B,** which drive the bulk of Black Friday volume — ensure inventory, staffing, and delivery capacity are prioritized there.
6. **Treat the ~0.5% high-value "big basket" transactions as a VIP segment —** understand what they're buying and consider a light loyalty/rewards nudge to encourage repeat big-basket behavior rather than treating them as outliers to ignore.
7. **Re-run this confidence-interval analysis periodically (e.g., each Black Friday season) —** the underlying gaps (Gender, Age) should be monitored over time; if the Gender gap narrows or the Marital-status gap opens up, targeting strategy should adapt accordingly.

Caveats

- This is Black-Friday-specific transactional data; the conclusions describe **Black Friday shopping behavior** and may not generalize to non-holiday shopping patterns.
- **Gender** here reflects the account/transaction record, which may not always represent who physically made or benefited from the purchase (e.g., gift shopping, shared household accounts).